# DSA 8020 Project 2

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## Contents

# Problem 1: Regression Modeling - Generalized Linear Model vs Linear Regression Model

Perform a detailed regression analysis using the gala data set. Use *Species* as the response variable, *excluding Endemics from this analysis*, to address the following questions:

```r
# loading data
data(gala)

# remove endemics from dataset
gala <- subset(gala, select = -Endemics)
summary(gala)
```

```
##     Species            Area             Elevation          Nearest
##  Min.   :  2.00   Min.   :   0.010   Min.   :  25.00   Min.   : 0.20
##  1st Qu.: 13.00   1st Qu.:   0.258   1st Qu.:  97.75   1st Qu.: 0.80
##  Median : 42.00   Median :   2.590   Median : 192.00   Median : 3.05
##  Mean   : 85.23   Mean   : 261.709   Mean   : 368.03   Mean   :10.06
##  3rd Qu.: 96.00   3rd Qu.:  59.237   3rd Qu.: 435.25   3rd Qu.:10.03
##  Max.   :444.00   Max.   :4669.320   Max.   :1707.00   Max.   :47.40
##      Scruz           Adjacent
##  Min.   :  0.00   Min.   :   0.03
##  1st Qu.: 11.03   1st Qu.:   0.52
##  Median : 46.65   Median :   2.59
##  Mean   : 56.98   Mean   : 261.10
##  3rd Qu.: 81.08   3rd Qu.:  59.24
##  Max.   :290.20   Max.   :4669.32
```

```r
head(gala)
```

```
##              Species  Area Elevation Nearest Scruz Adjacent
## Baltra            58 25.09       346     0.6   0.6     1.84
## Bartolome         31  1.24       109     0.6  26.3   572.33
## Caldwell           3  0.21       114     2.8  58.7     0.78
## Champion          25  0.10        46     1.9  47.4     0.18
## Coamano            2  0.05        77     1.9   1.9   903.82
## Daphne.Major      18  0.34       119     8.0   8.0     1.84
```

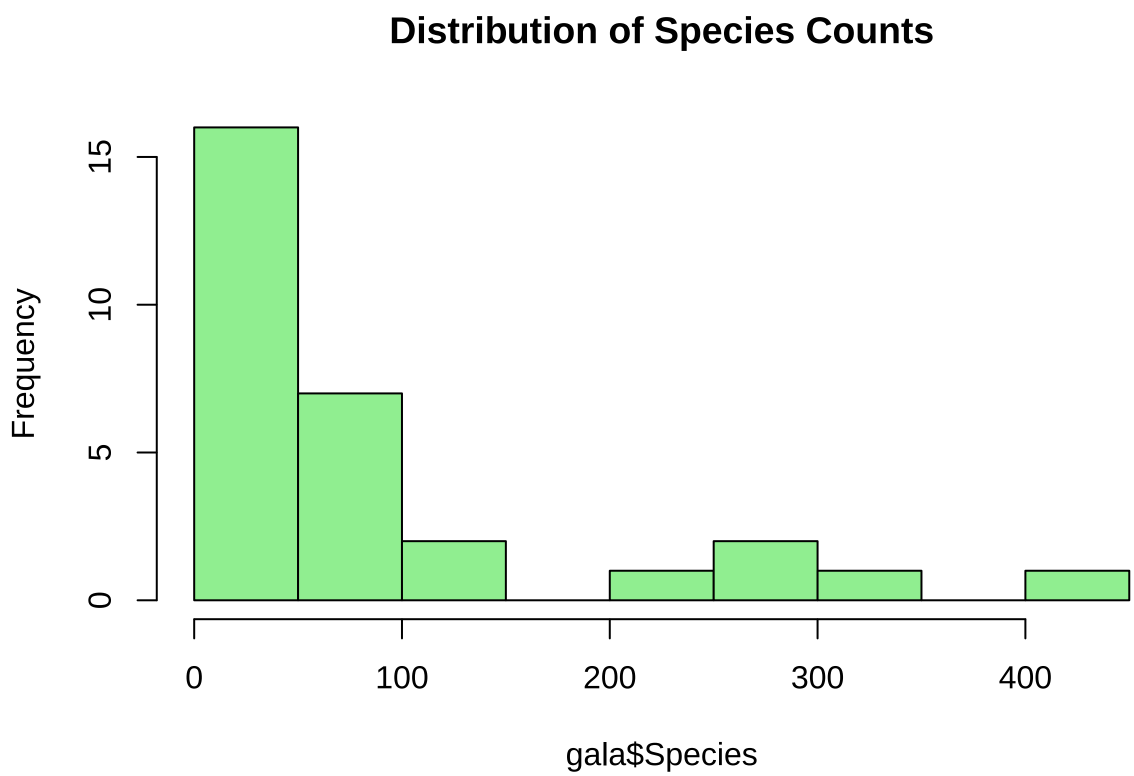
---

**1. Conduct a regression analysis using one of the generalized linear models we covered in Week 7 (i.e., logistic regression or Poisson regression) and perform model selection.**

- From the dataset description, we are told that 'Species' is a count of the number of plant species found on the island.
- Since 'Species' is a count, a normal distribution (Linear Regression) would not be the best choice as that allows for negative values.
- Species' is not a binary value, nor is it a category type variable, thus Logistic Regression is not a reasonable choice as our GLM.

- Using summary(gala) we can confirm that 'Species' has no negative values (min=2). Based on these details, we can predict that Poisson will be the best fit to the data. However, we will confirm this further…

```
# histogram of species
hist(gala$Species, breaks = 10, main = "Distribution of Species Counts", col = 'lightgreen')
```




- From the histogram, we can see that the distribution of 'Species' is skewed right, alluding to Poisson model being the best choice for our generalized linear model. However we must check for over-dispersion as if true then a quasi-poisson or negative binomial model would be a better choice.

```
# mean vs varience
mean(gala$Species)
```

```
## [1] 85.23333
```

```
var(gala$Species)
```

```
## [1] 13140.74
```

- We can see that the variance of 'Species' is greater than it's mean, indicating that Poisson (including quasi-poisson due to the **significantly** large difference) would not be the best fit due to over-dispersion. Rather, a negative binomial model would be better.

```
# fitting neg binom
gala_negbinom <- glm.nb(Species ~ Area + Elevation + Nearest + Scruz + Adjacent, data = gala)
summary(gala_negbinom)
```

```
##
## Call:
## glm.nb(formula = Species ~ Area + Elevation + Nearest + Scruz +
##     Adjacent, data = gala, init.theta = 1.674602286, link = log)
##
## Coefficients:
##               Estimate Std. Error z value Pr(>|z|)
## (Intercept)  2.9065247  0.2510344  11.578  < 2e-16 ***
## Area        -0.0006336  0.0002865  -2.211 0.027009 *
## Elevation    0.0038551  0.0006916   5.574 2.49e-08 ***
## Nearest      0.0028264  0.0136618   0.207 0.836100
## Scruz       -0.0018976  0.0028096  -0.675 0.499426
## Adjacent    -0.0007605  0.0002278  -3.338 0.000842 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for Negative Binomial(1.6746) family taken to be 1)
##
##     Null deviance: 88.431  on 29  degrees of freedom
## Residual deviance: 33.196  on 24  degrees of freedom
## AIC: 304.22
##
## Number of Fisher Scoring iterations: 1
##
##
##               Theta:  1.675
##           Std. Err.:  0.442
##
##  2 x log-likelihood:  -290.223
```

- Using a negative binomial model as our Generalized Linear Model, we can see that certain variables (such as 'Nearest' & 'Scruz') are not useful as predictors. I will now optimize our GLM and refine the model using stepwise model selection.

```r
# optimizing neg binom stepwise
# dir='backward' less risk of overfit than 'both'
optim_galaNB <- step(gala_negbinom, direction = "backward")
```

```
## Start:  AIC=302.22
## Species ~ Area + Elevation + Nearest + Scruz + Adjacent

## Warning: glm.fit: algorithm did not converge

##             Df Deviance    AIC
## - Nearest    1   33.263 300.29
## - Scruz      1   33.569 300.60
## <none>           33.196 302.22
## - Area       1   36.665 303.69
## - Adjacent   1   41.151 308.18
## - Elevation  1   65.866 332.89
##
## Step:  AIC=300.29
## Species ~ Area + Elevation + Scruz + Adjacent
```

```
## Warning: glm.fit: algorithm did not converge

##             Df Deviance    AIC
## - Scruz      1   33.500 298.60
## <none>           33.194 300.29
## - Area       1   36.796 301.89
## - Adjacent   1   41.563 306.66
## - Elevation  1   67.553 332.65
##
## Step:  AIC=298.59
## Species ~ Area + Elevation + Adjacent

## Warning: glm.fit: algorithm did not converge

##             Df Deviance    AIC
## <none>           33.155 298.59
## - Area       1   36.676 300.11
## - Adjacent   1   41.723 305.16
## - Elevation  1   67.847 331.29
```

```
summary(optim_galaNB)
```

```
##
## Call:
## glm.nb(formula = Species ~ Area + Elevation + Adjacent, data = gala,
##     init.theta = 1.651523202, link = log)
##
## Coefficients:
##               Estimate Std. Error z value Pr(>|z|)
## (Intercept)  2.8149002  0.2231453  12.615  < 2e-16 ***
## Area        -0.0006449  0.0002804  -2.300 0.021459 *
## Elevation    0.0039299  0.0006761   5.812 6.16e-09 ***
## Adjacent    -0.0007943  0.0002196  -3.616 0.000299 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for Negative Binomial(1.6515) family taken to be 1)
##
##     Null deviance: 87.279  on 29  degrees of freedom
## Residual deviance: 33.155  on 26  degrees of freedom
## AIC: 300.59
##
## Number of Fisher Scoring iterations: 1
##
##
##               Theta:  1.652
##           Std. Err.:  0.434
##
##  2 x log-likelihood:  -290.593
```

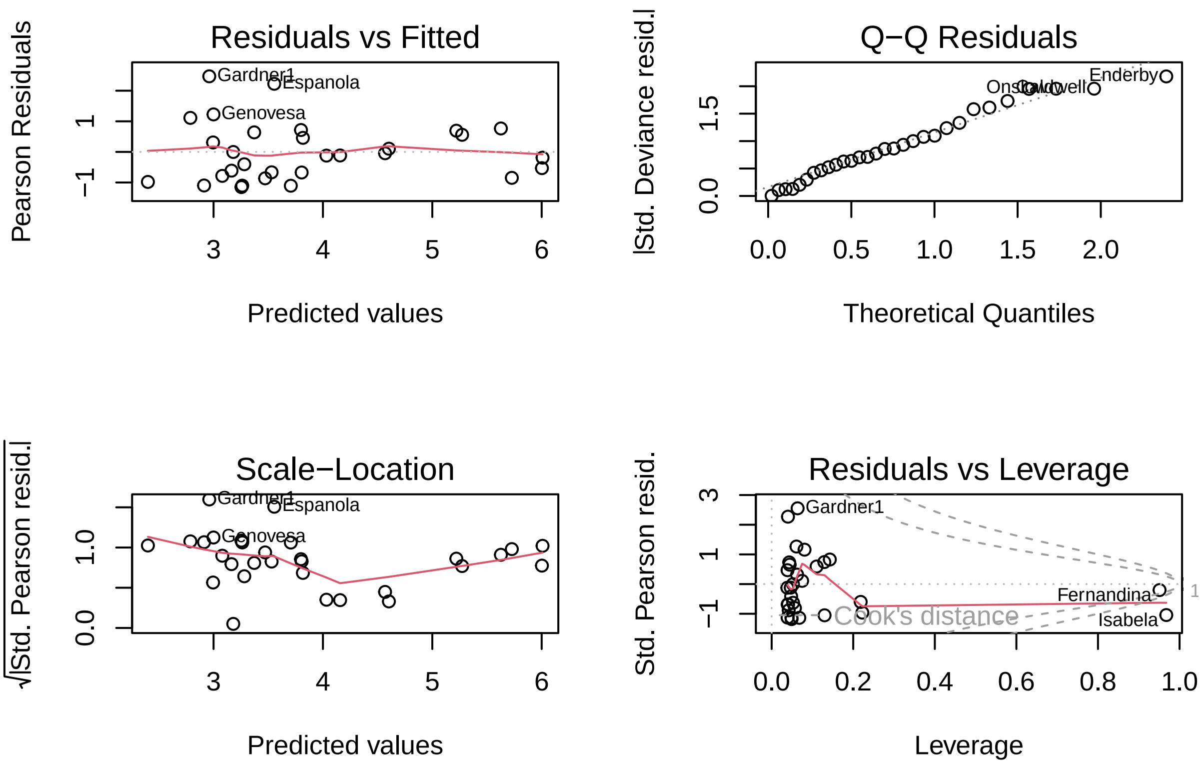
- As predicted, we can safely remove 'Nearest' & 'Scruz'. Based on this, we get a final regression equation of…

Species ~ Area + Elevation + Adjacent

- We will not ensure that our model is valid by checking if model assumption are met:

```
par(mfrow = c(2, 2))
plot(optim_galaNB)
```

```
## Warning in sqrt(crit * p * (1 - hh)/hh): NaNs produced
## Warning in sqrt(crit * p * (1 - hh)/hh): NaNs produced
```




- Check for model assumptions:
  - Residual vs. Fitted: Residuals are randomly scattered around 0 (x-axis) and do not show a clear pattern, confirming assumptions of homoscedasticity and linearity are met.
  - QQ Residuals: Residuals fall close to diagonal, confirms residuals follow a normal distribution.

**2. Perform a linear regression analysis and conduct model selection again to determine the most appropriate linear regression model. Compare it with the GLM model selected in the previous question and comment on your findings.**

```
# basic linear model
lm_gala <- lm(Species ~ ., data = gala)

# stepwise model selection
optim_galaLM <- step(lm_gala, direction = "backward")
```

```
## Start:  AIC=251.93
## Species ~ Area + Elevation + Nearest + Scruz + Adjacent
##
##             Df Sum of Sq    RSS    AIC
## - Nearest    1         0  89232 249.93
## - Area       1      4238  93469 251.33
## - Scruz      1      4636  93867 251.45
## <none>                    89231 251.93
## - Adjacent   1     66406 155638 266.62
## - Elevation  1    131767 220998 277.14
##
## Step:  AIC=249.93
## Species ~ Area + Elevation + Scruz + Adjacent
##
##             Df Sum of Sq    RSS    AIC
## - Area       1      4436  93667 249.39
## <none>                    89232 249.93
## - Scruz      1      7544  96776 250.37
## - Adjacent   1     72312 161544 265.74
## - Elevation  1    139445 228677 276.17
##
## Step:  AIC=249.39
## Species ~ Elevation + Scruz + Adjacent
##
##             Df Sum of Sq    RSS    AIC
## - Scruz      1      6336 100003 249.35
## <none>                    93667 249.39
## - Adjacent   1     69860 163527 264.11
## - Elevation  1    275784 369451 288.56
##
## Step:  AIC=249.35
## Species ~ Elevation + Adjacent
##
##             Df Sum of Sq    RSS    AIC
## <none>                   100003 249.35
## - Adjacent   1     73251 173254 263.84
## - Elevation  1    280817 380820 287.47
```

```
summary(optim_galaLM)
```

```
##
## Call:
## lm(formula = Species ~ Elevation + Adjacent, data = gala)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -103.41  -34.33  -11.43   22.57  203.65
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1.43287   15.02469   0.095 0.924727
## Elevation    0.27657    0.03176   8.707 2.53e-09 ***
## Adjacent    -0.06889    0.01549  -4.447 0.000134 ***
## ---
```

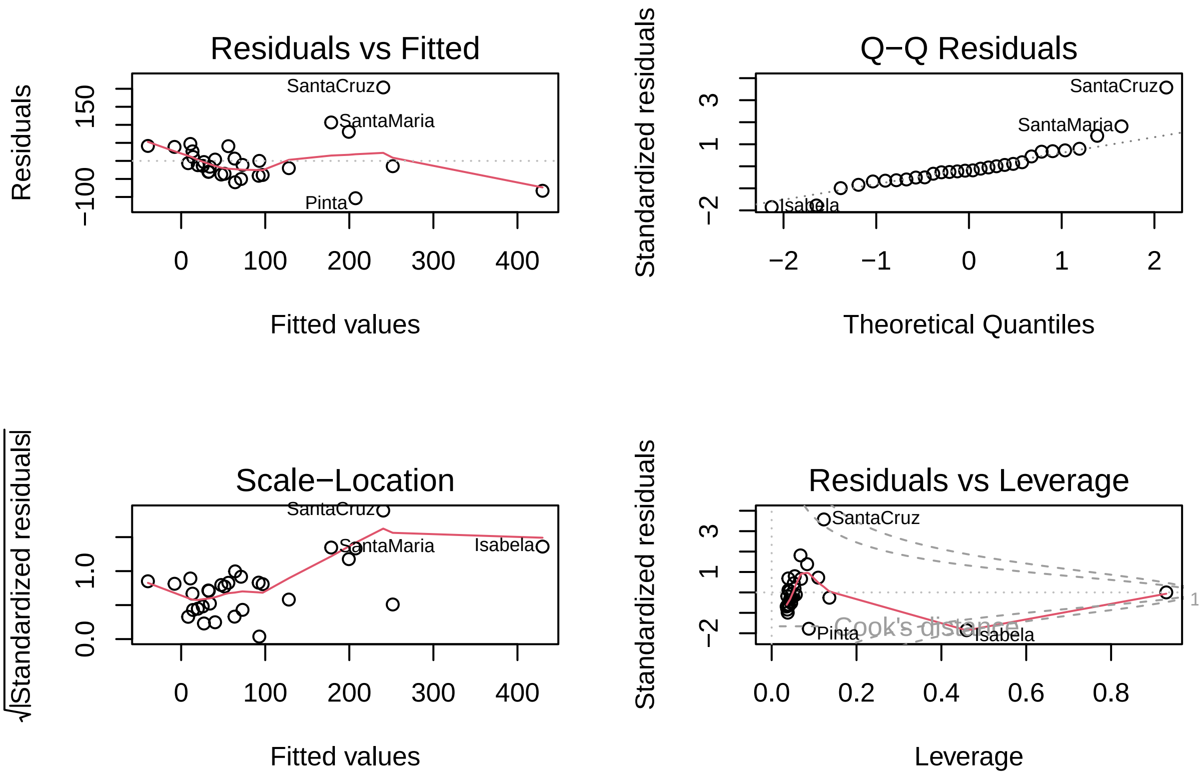
```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 60.86 on 27 degrees of freedom
## Multiple R-squared:  0.7376, Adjusted R-squared:  0.7181
## F-statistic: 37.94 on 2 and 27 DF,  p-value: 1.434e-08
```

- Similar to our GLM, stepwise model selection excluded predictors 'Nearest' & 'Scruz'. Additionally, stepwise selection also eliminated predictor 'Area'. This gives us a regression equation of:

  Species ~ Elevation + Adjacent

- We will now check model assumptions to ensure optimized linear model is valid.

```r
par(mfrow = c(2, 2))
plot(optim_galaLM)
```




```
- Residual vs. Fitted: No clear pattern and located near 0, confirms homoscedasticity and linearity assu

- QQ Plot: Residuals lie along diagonal, confirms normal distribution.
```

- Now we will compare our linear model with our previous GLM (negative binomial).

```r
AIC(optim_galaLM, optim_galaNB)
```

```
##              df      AIC
## optim_galaLM  4 336.4891
## optim_galaNB  5 300.5934
```

```r
BIC(optim_galaLM, optim_galaNB)
```

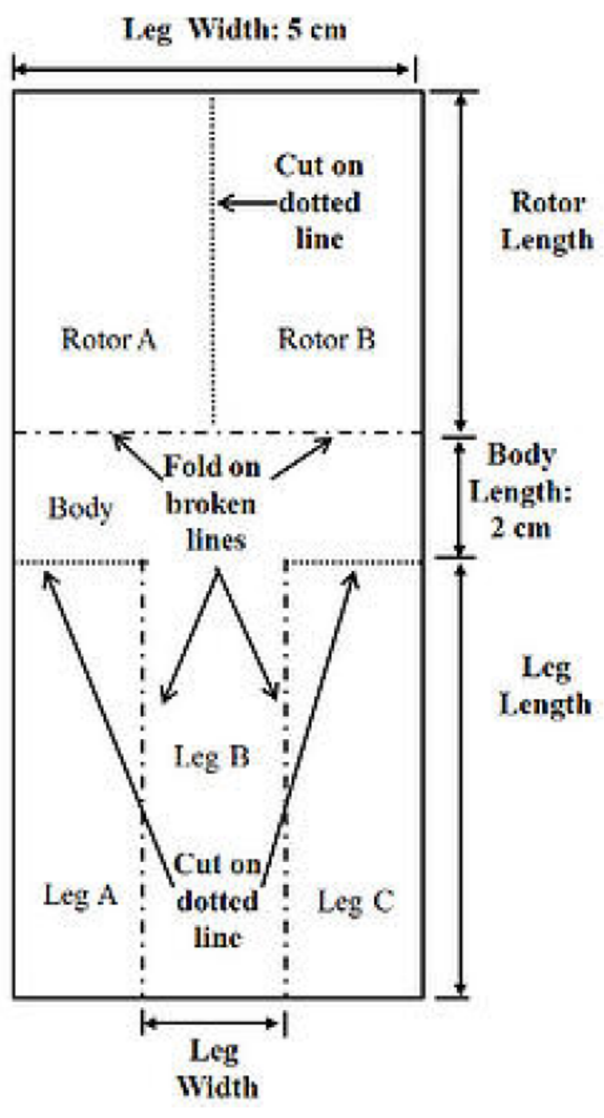
```
##              df      BIC
## optim_galaLM  4 342.0938
## optim_galaNB  5 307.5994
```

- Our negative binomial GLM shows to have lower AIC and BIC criterion value, proving it is the better choice over our linear regression model.

---

## Problem 2: Paper Helicopter Experiment

A researcher would like to investigate how long a paper 'helicopter' can stay in the air when dropped from a height of 2 meters. Here's how she makes the paper helicopters to be used for the experiment.




Source: https://blog.minitab.com/en/learning-design-of-experiments-with-paper-helicopters-and-minitab

- Step 1: Cut the paper to a width of 5cm.
- Step 2: Cut the paper the length of paper rotor length plus leg length, and add 2 cm for the body.
- Step 3: Cut dotted lines at Leg A and Leg C. The length of each cut is 5 cm minus leg width divided by 2.
- Step 4: Fold legs A and B onto leg B.
- Step 5: Fold rotor A and rotor B in opposite directions. They should form 90° to the body and be 180° away from each other.

Some potential treatment factors are:

```
1) Paper type (light, medium, and heavy)

2) Rotor length (7.5cm or 8.5cm)

3) Leg length (7.5cm or 12cm)

4) Leg width (3.2cm or 5cm).
```

Answer the following questions:

**A. Suppose the researcher would like to compare the responses between different paper types while keeping other factors fixed. What are the experimental units and measurement units [1] in this study? Which design can she apply?**

**Answer:**

- Experimental Unit: Our helicopter which will undergo the experiment, with different treatments or *paper type* being applied to them. *paper type* must be one of the following values: light, medium, or heavy.
- Measurement Unit: *flight time*, likely measured in seconds or milliseconds. The researcher would like to find the helicopter with the maximum *flight time*, with each helicopter receiving different treatments (*paper type*).
- Optimal Experiment Design: *Completely Randomized Design (CRD)*, since the researcher is observing response for only **one** changing factor (*paper type*) in each experimental unit (helicopter), while leaving other factors constant.

**B. Suppose the researcher would like to conduct an experiment in (A) using 15 paper helicopters. Explain how she can perform the randomization and replication procedures here.**

**Answer:**

- Randomization: Since only one factor, *paper type*, is changing for each treatment in the CRD, the researcher should assign each helicopter randomly to a paper type. This can be accomplished using a pseudo-random number generation in range(0, 2), with each *paper type* (light, medium, or heavy) being determined by the output. The **sample()** function in R or with Python's **random** package can accomplish this.

[1] Please don't confuse it with the other experimental unit, for example, seconds. Check the lecture slides in Week 8 for the definition of experimental unit in the context of design of experiments.

- Replication: Since the researcher will only produce a total of 15 experimental units (helicopters), and there are 3 types of treatments (*paper type*), each treatment (*paper type*) should make up exactly 5 helicopters. This is necessary to not only account for variability, but also to ensure enough data points are present for ANOVA statistical analysis to be valid.

**C. Think of a situation in which the researcher would need to consider using a randomized complete block design (RCBD).**

**Answer:**

- Randomized CRD will be useful when noise is observed to introduce variability in the experiment. RCRD will reduce this experimental error within blocks (i.e. testing instances), resulting in more precise comparison between our changing treatments of *paper type*.
- For example, if the researcher performs blocks at different times in the day, noise variables such as wind, temperature, and other factors from the environment could impact the results - preventing proper comparison between the treatments alone. For each block, the researcher should conduct experiment on all three treatments simultaneously from the same location, to ensure that noise variables are not causing variability in *flight time*, rather, just the treatment.

**D. If the researcher wants to study the effects of leg length and width on the flying time of heavy paper helicopters, which design is most appropriate for this situation? If the researcher wants to have 5 replicates for each treatment combination, how many heavy paper helicopters does she need?**

**Answer:**

- In this new experiment setup, the helicopter remains our experimental unit. Now, *paper type* is statically 'heavy', and its rotor length is kept constant; our treatments will be composed of variation of *leg length* and *leg width*. Our measurement unit will still remain *flight time*.
- Since the researcher wants to study the effects of 2 different treatments (changes in leg length (1) and width (2)), and each of those 2 treatments can be one of 2 values (leg length: 7.5 or 12cm; leg width: 3.2 or 5cm), we can see that we would use a **2x2** design. Thus a *Factorial Design* for the experiment would be optimal.
  - Factorial design will allow to measure both the main effects (does leg lenght or width affect flight time?) and the interaction effects (does effect of leg lenght depend on leg width, and vice versa).
  - Will have a total of 4 possible total treatment combinations.
- If 5 replicates (helicopters) per treatment are required, then the researcher would need a total of 20 'heavy' *paper type* helicopters, all of the same *rotor length*.

**E. Conduct your own experiment, including designing the experiment, collecting and analyzing the data, to investigate the effects of rotor length, leg length, and leg width on the response, while keeping the paper type fixed.**

---

## Problem 3: Paper Helicopter Computer Experiment

A sophisticated computer model has been developed to study how the flying time depends on the rotor length and leg length. The output of the computer can be found in ProjectII_prob2_data.RData.

**x1**: rotor length (cm)

**x2**: leg length (cm)

**y**: flying time (seconds)

```
# loading data
load("ProjectII_prob2_data.RData")
```

Answer the following questions:

#### A: Compute and visualize the prediction surface by computing the predicted values at a dense set of grid points:

x1g <- seq(6, 10, 0.1)

x2g <- seq(6, 12, 0.2)

grids <- expand.grid(x1g, x2g)

```
copter_df <- data.frame(x1 = x1, x2 = x2, y = y)

# fitting GAM regression
copter_gam <- gam(y ~ x1 + x2, data = copter_df)
summary(copter_gam)
```

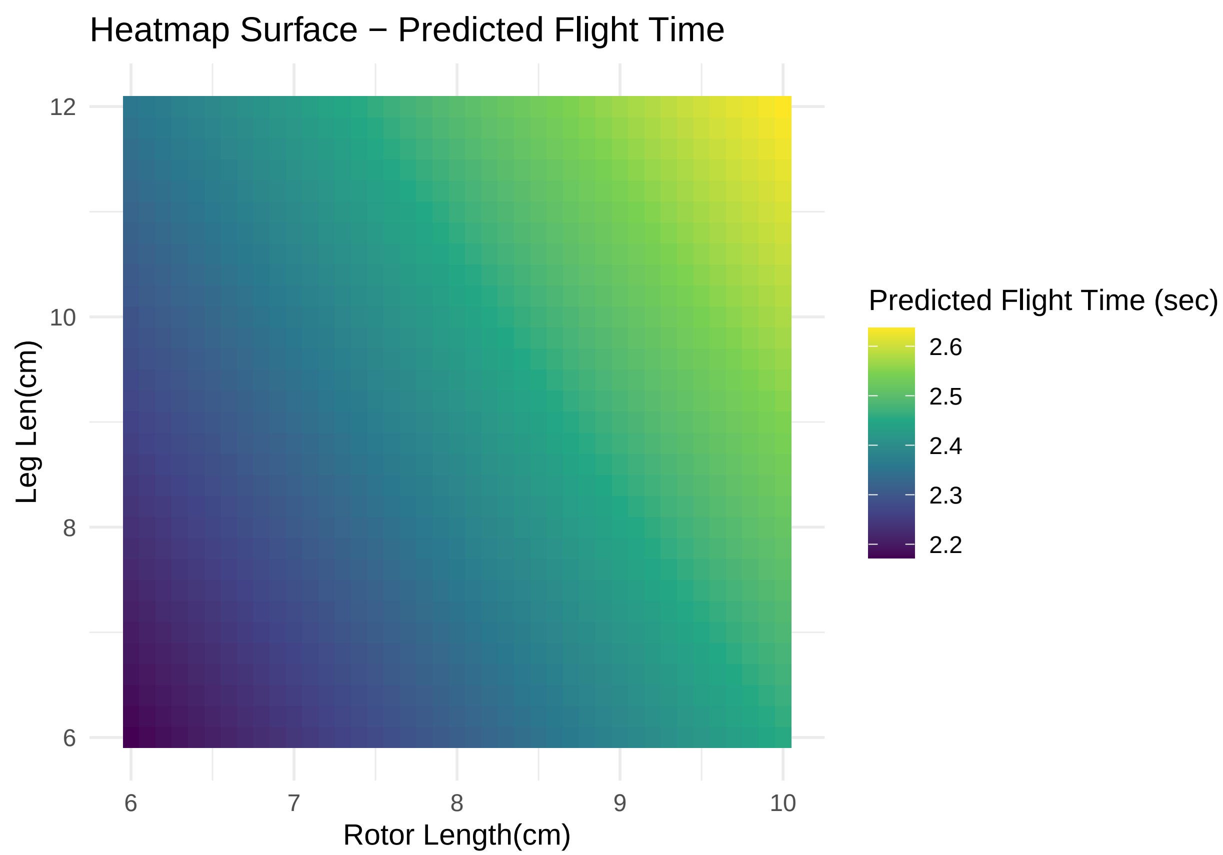
```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## y ~ x1 + x2
##
## Parametric coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 1.567685   0.035622   44.01   <2e-16 ***
## x1          0.070438   0.003581   19.67   <2e-16 ***
## x2          0.030348   0.002389   12.70   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## R-sq.(adj) =   0.85   Deviance explained = 85.3%
## GCV = 0.0017591  Scale est. = 0.0017064  n = 100
```

```
# dense set
x1g <- seq(6, 10, 0.1)
x2g <- seq(6, 12, 0.2)
```

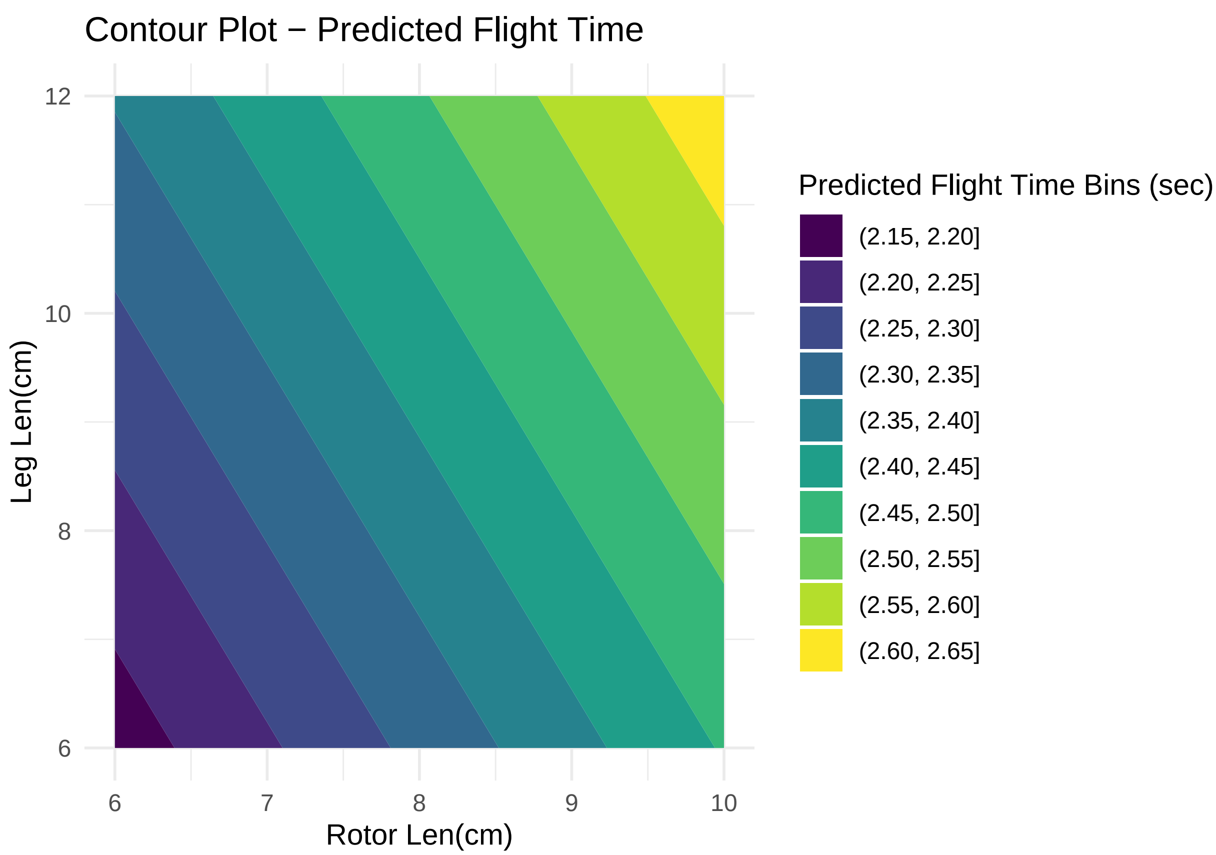
```r
grids <- expand.grid(x1 = x1g, x2 = x2g)

# pred on grid
grids$pred_y <- predict(copter_gam, newdata = grids)

# visual pred surface heatmap
ggplot(grids, aes(x = x1, y = x2, fill = pred_y)) +
  scale_fill_viridis_c() + # match contour plt color
  geom_tile() +
  labs(title = "Heatmap Surface - Predicted Flight Time",
       x = "Rotor Length(cm)",
       y = "Leg Len(cm)",
       fill = "Predicted Flight Time (sec)") +
  theme_minimal()
```




```r
# contour
ggplot(grids, aes(x = x1, y = x2, z = pred_y)) +
  geom_contour_filled() +
  labs(title = "Contour Plot - Predicted Flight Time",
       x = "Rotor Len(cm)",
       y = "Leg Len(cm)",
       fill = "Predicted Flight Time Bins (sec)") +
  theme_minimal()
```




**B. Compute and visualize the prediction uncertainty using a Gaussian process model. Briefly comment on your findings.**

```
# fitting gaussian model
copter_GPFit <- mlegp(copter_df[, 1:2], copter_df[, 3])
```

```
## no reps detected - nugget will not be estimated
##
## ========== FITTING GP # 1 ==============================
## running simplex # 1...
## ...done
## ...simplex #1 complete, loglike = 222.252440 (convergence)
## running simplex # 2...
## ...done
## ...simplex #2 complete, loglike = 222.252440 (convergence)
## running simplex # 3...
## ...done
## ...simplex #3 complete, loglike = 222.252439 (convergence)
## running simplex # 4...
## ...done
## ...simplex #4 complete, loglike = 222.252440 (convergence)
## running simplex # 5...
## ...done
## ...simplex #5 complete, loglike = 222.252440 (convergence)
```

```
##
## using L-BFGS method from simplex #1...
## ...L-BFGS method complete
##
## Maximum likelihood estimates found, log like =  222.252440
## creating gp object......done
```

```r
summary(copter_GPFit)
```

```
##
## Total observations = 100
## Dimensions = 2
##
## mu = 2.396247
## sig2:    0.005812706
## nugget:  0
##
## Correlation parameters:
##
##       beta a
## 1 1.753702 2
## 2 1.139921 2
##
## Log likelihood = 222.2524
##
## CV RMSE: 0.01769931
## CV RMaxSE: 0.006604805
```

```r
# make preds on dense grid
# y_hats <- predict(copter_GPFit, grids, se.fit = TRUE)
# print(y_hats)
```